COLLEGE OF ENGINEERING GUINDY CAMPUS
ANNA UNIVERSITY, CHENNAI

DEPARTMENT OF INFORMATION SCIENCE AND TECHNOLOGY (BATCH 1)

DATA STRUCTURE PROJECT

HOSTEL COMPLAINT PRIORITY MANAGEMENT SYSTEM USING DATA STRUCTURES IN C

SUBMITTED BY:

KAMINI (2025115051)

SUBHEKSHA K.S (2025115109)

ABSTRACT

This project presents a menu-driven Hostel Complaint Priority Management System developed in C. Hostel students may face maintenance issues such as water supply problems, electrical faults, fan repairs and plumbing issues. Manual complaint handling can make tracking and prioritising complaints difficult.

The program provides options to register a complaint, display all complaints, search by complaint ID, update complaint status and exit. A complaint record contains an ID, student name, room number, category, priority and status. The source code defines a Complaint structure, a linked-list pointer and a data file named complaints.txt. Priority levels are Low, Medium and High. The system aims to make complaint recording and tracking more organised.

1. INTRODUCTION

Hostel maintenance is important for students' safety, comfort and daily activities. If complaints are recorded only in notebooks or communicated informally, information may be missed or delayed. This project demonstrates a simple console-based system for recording and tracking hostel maintenance complaints using C and data-structure concepts.

2. PROBLEM STATEMENT

Manual complaint handling can make it difficult to maintain records, find a particular complaint, identify urgency and track whether an issue is pending, in progress or resolved. A computer-based system is needed to record complaint details, assign a priority level and retrieve or update complaint information.

3. OBJECTIVES

• Register hostel maintenance complaints with student and room details.

• Assign a priority level to each complaint.

• Generate a unique complaint ID for tracking.

• Display recorded complaints.

• Search for a complaint using its ID.

• Update a complaint's status.

• Demonstrate structures, pointers, linked lists and file handling in C.

4. SCOPE OF THE PROJECT

The demonstrated menu includes Register Complaint, Display All Complaints, Search Complaint by ID, Update Complaint Status and Exit. The project focuses on basic complaint recording and tracking through a console interface. It is an academic demonstration, not a complete hostel administration platform.

Features such as login authentication, notifications, a graphical interface, advanced analytics and online complaint submission are possible future enhancements.

5. DATA STRUCTURES AND CONCEPTS USED

5.1 Structure: A C structure groups fields for a complaint, including ID, student name, room number, category, priority and status.

5.2 Linked List: The source code defines a complaint node with a pointer to the next complaint. A linked list can connect complaint records dynamically.

5.3 Pointers: Pointers connect nodes and support traversal through the linked list.

5.4 File Handling: The program defines complaints.txt as its data file. Test the final program by restarting it to confirm whether records persist between runs.

5.5 Priority Levels: The program prompts for priority 1 = Low, 2 = Medium and 3 = High.

Note: Queue-based processing, automatic sorting and report generation should only be described as implemented if they are present and tested in the final C source code.

6. SYSTEM REQUIREMENTS

Hardware: Laptop or desktop computer.

Software: Windows, a C compiler such as GCC/MinGW, and a text editor or IDE.

Programming language: C.

Data file: complaints.txt, as defined in the source code.

7. SYSTEM DESIGN AND WORKING

1. Start the program and display the main menu.

2. Read the user’s menu choice.

3. Register Complaint: collect student name, room number, category and priority, then assign/display a complaint ID.

4. Display All Complaints: show the complaint records available to the program.

5. Search Complaint by ID: request an ID and display the matching record if found.

6. Update Complaint Status: request the complaint ID and new status, then update the record if found.

7. Exit: terminate the program.

8. Repeat the menu after an operation until the user chooses Exit.

8. ALGORITHM

1. Start.

2. Initialise the complaint list and complaint ID counter.

3. Display the menu and read the choice.

4. Perform the selected operation: register, display, search, update status or exit.

5. For registration, read complaint details, assign an ID and store the record.

6. For search or update, locate the complaint using its ID.

7. Display a suitable result message.

8. Return to the menu unless Exit was selected.

9. Stop.

9. IMPLEMENTED MENU OPTIONS

Option 1 — Register Complaint: collects student name, room number, category and priority, then displays a generated complaint ID.

Option 2 — Display All Complaints: displays complaint records available to the program.

Option 3 — Search Complaint by ID: searches using the complaint's unique ID.

Option 4 — Update Complaint Status: changes the selected complaint's status using the options offered by the program, such as Pending, In Progress or Resolved.

Option 5 — Exit: ends the program.

10. TESTING AND RESULTS

The following are suggested test cases. Run each one on the final source code and confirm the actual result before submission.

Test	Action	Expected result
1	Register complaint with valid details	A complaint is registered and an ID is displayed.
2	Display all complaints	Available complaint records are displayed.
3	Search with an existing ID	The matching complaint is displayed.
4	Search with a non-existing ID	A not-found message is displayed.
5	Update an existing complaint	The selected complaint status is updated.
6	Exit	The program terminates.

11. LIMITATIONS

• The current application uses a console interface.

• It depends on valid user input and local storage.

• A web/mobile interface and user authentication are not demonstrated.

• Record persistence should be verified by restarting the application.

• Sorting, queue processing and summary report generation should not be claimed unless verified in the code.

12. FUTURE ENHANCEMENTS

• Add administrator login and access control.

• Display complaints in priority order.

• Add filtering by category, priority and status.

• Generate summaries of pending, in-progress and resolved complaints.

• Improve input validation and error messages.

• Develop a graphical or web-based interface.

• Notify students when complaint status changes.

13. CONCLUSION

The Hostel Complaint Priority Management System demonstrates a basic way to organise hostel maintenance complaints using C. It provides menu options for registering complaints, displaying records, searching by complaint ID and updating complaint status. The project applies structures, pointers, linked-list concepts and file handling to a practical hostel scenario. Further testing and enhancements can improve reliability and usefulness.